Definition 1. ***Simplex*** *Given n linearly independent vectors connect each of the end points to get n+1 verticies.*

Given a simplex in $\mathbb{R}^n$ let $x^{(0)}, x^{(1)}, \ldots, x^{(n)}$ denote the verticies. Let $y_i = f(x^{(i)})$ for $i = 1 \ldots n$. Denote y_{iz} as the farthest away from the minimum, y_{iy} as the second farthest, and y_{ia} is the closest to the minimum. Notice the indicies may not be uniquely determined, but we want $ia \neq iy,\ iy \neq iz,$ and $iz \neq ia$. We call $y_{ia} = \min_i y_i$, $y_{iz} = \max_i y_i$, $y_{iy} = \max_{i \neq z} y_i$.

Definition 2. *centroid denoted as* $\hat{x} = \frac{1}{n}\sum_{i=0}^{n} x^{(i)}$ *for* $i \neq iz$.

We denote $x^{(r)}$ as the the reflextion and $x^{(e)}$ as the extension.